

Gated 5G simulation scaling from 2 to 12 nodes

384

Final shards

Hash-validated across 12 nodes

90.9%

CPU efficiency

Aggregate final-campaign utilization

85.4 min

12-node wall time

Complete Static / Reactive / MPC matrix

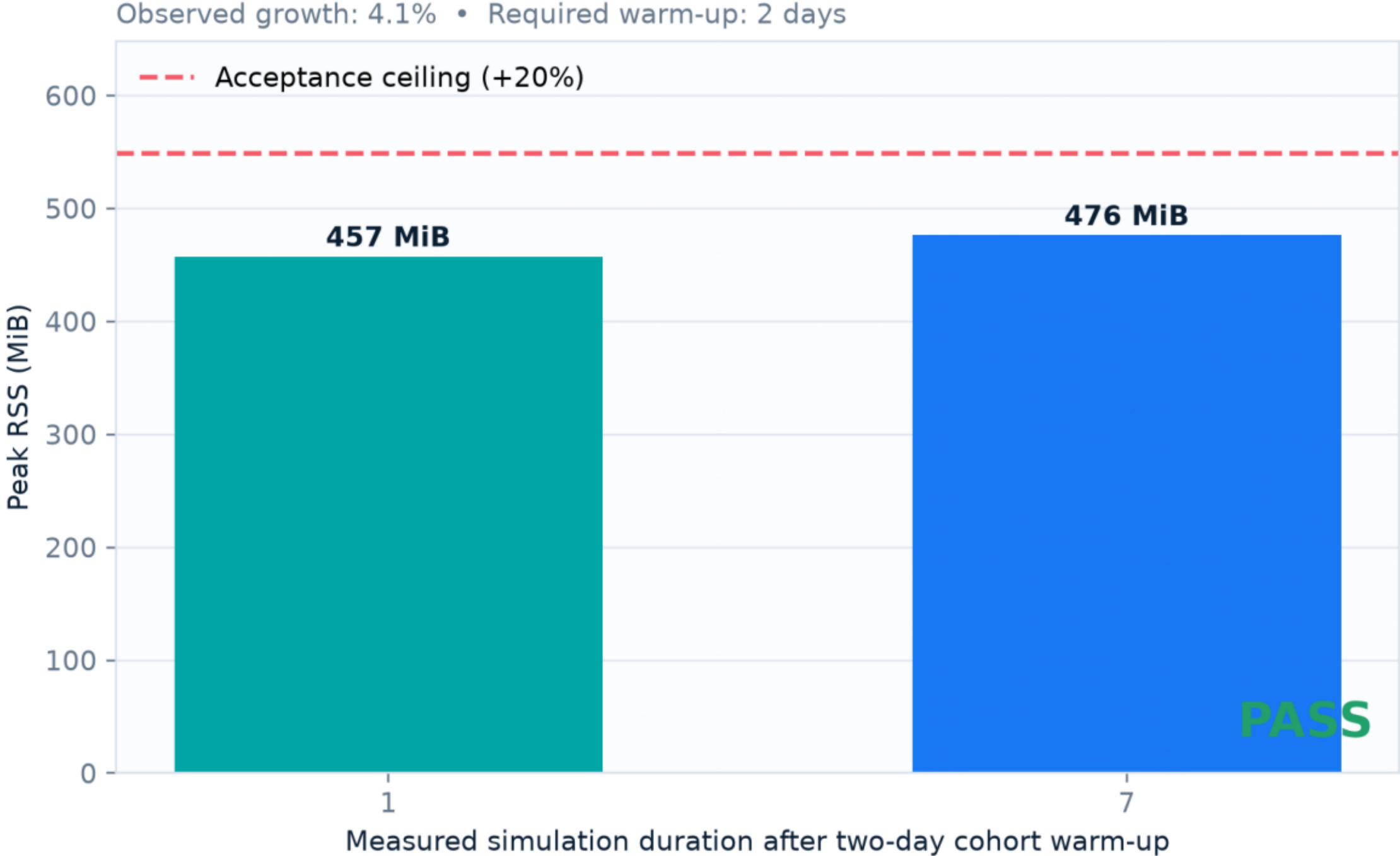
0

Worker failures

Zero swap across every gated phase

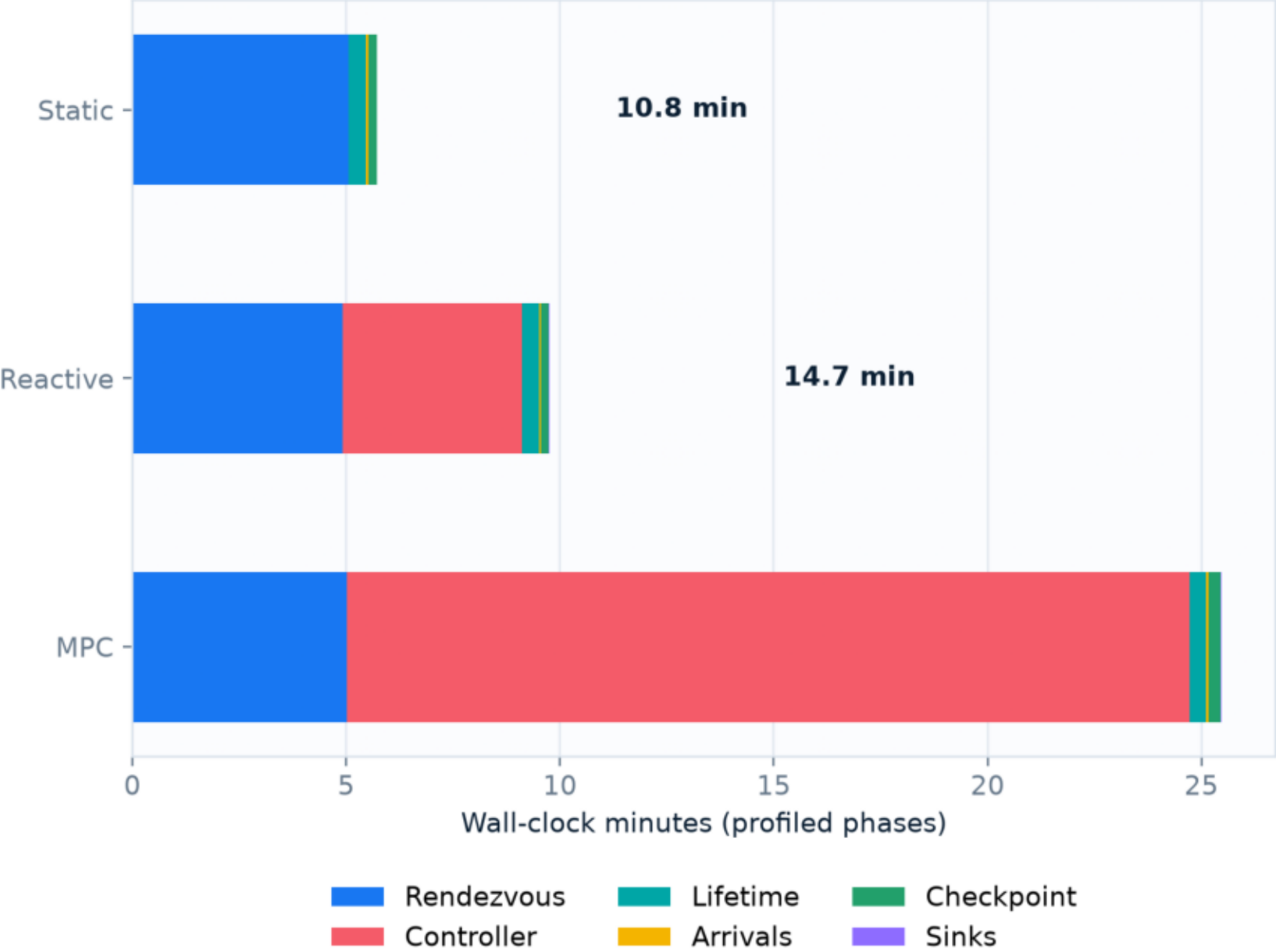
All production gates PASSED: 2 → 4 → 12 nodes, frozen inputs, zero worker/establishment failures, and zero swap.

Streaming memory remains flat as duration grows 7×

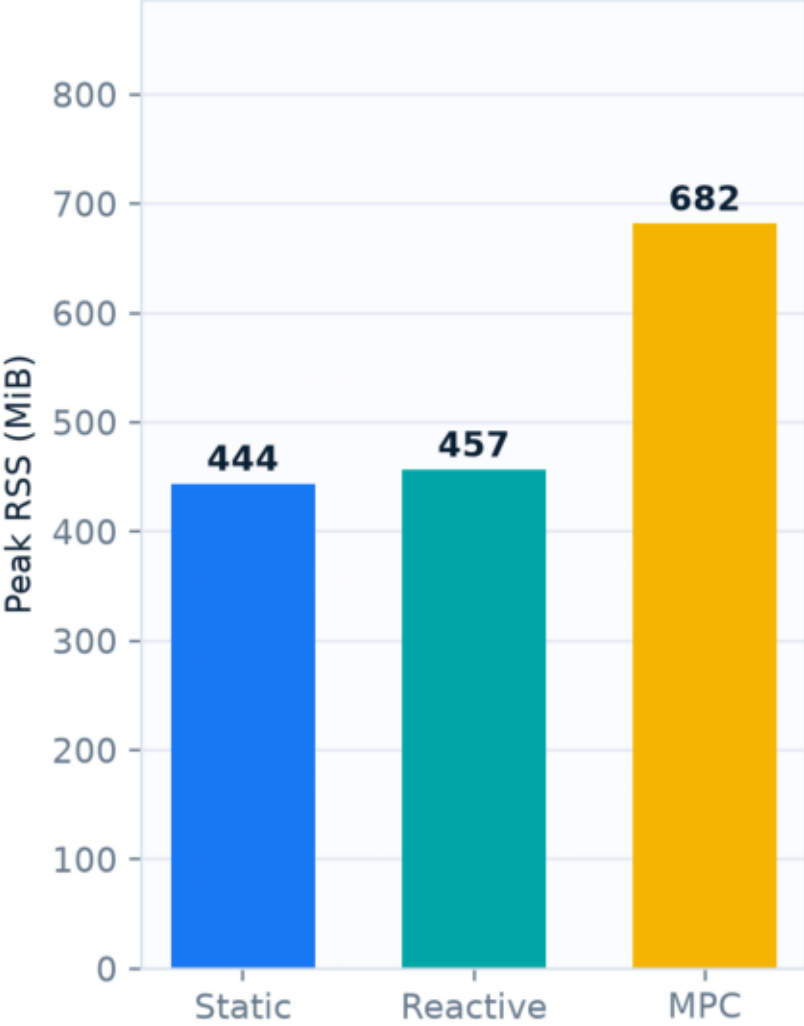


Controller characterization on the C-DOT node

Where one extreme day spends its time

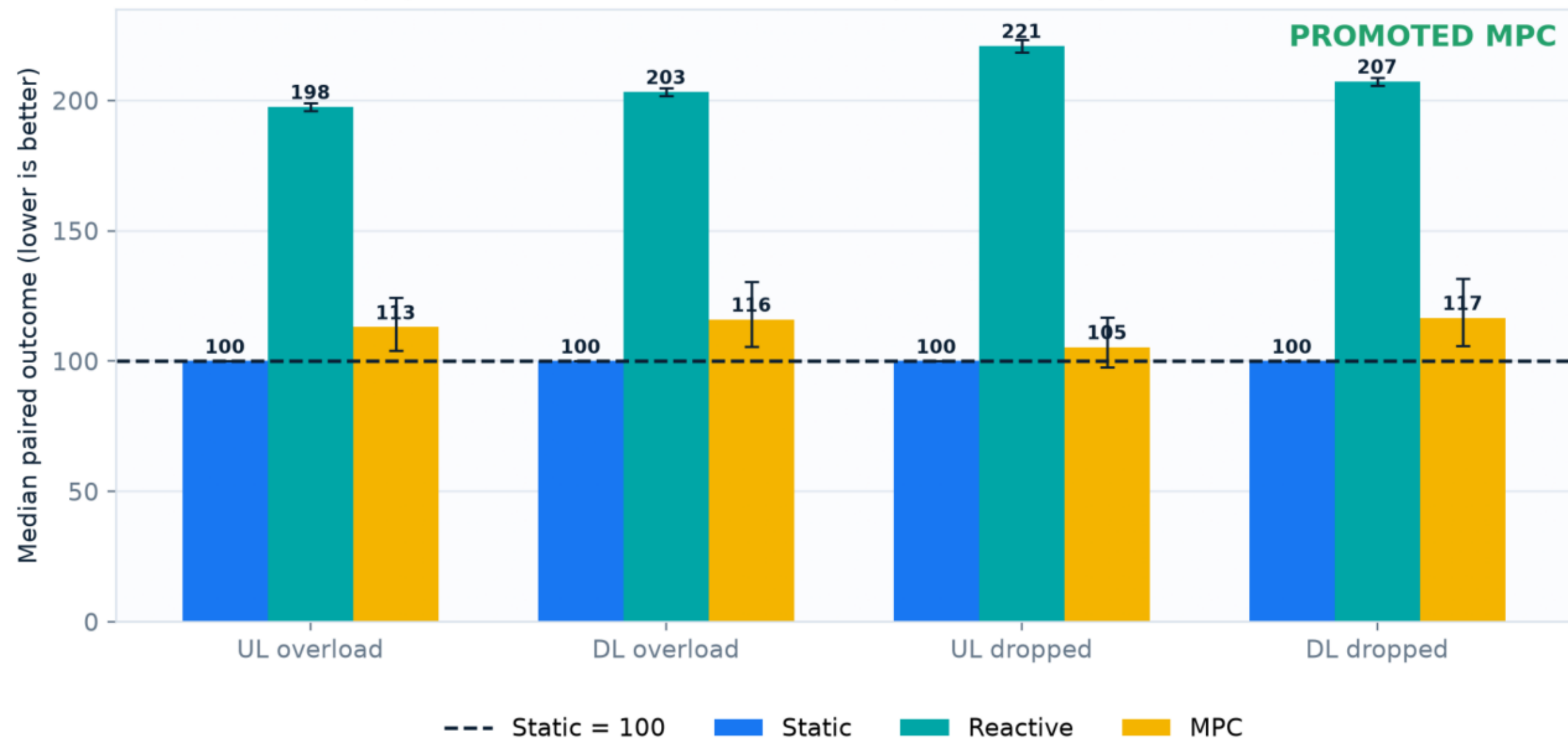


Peak memory

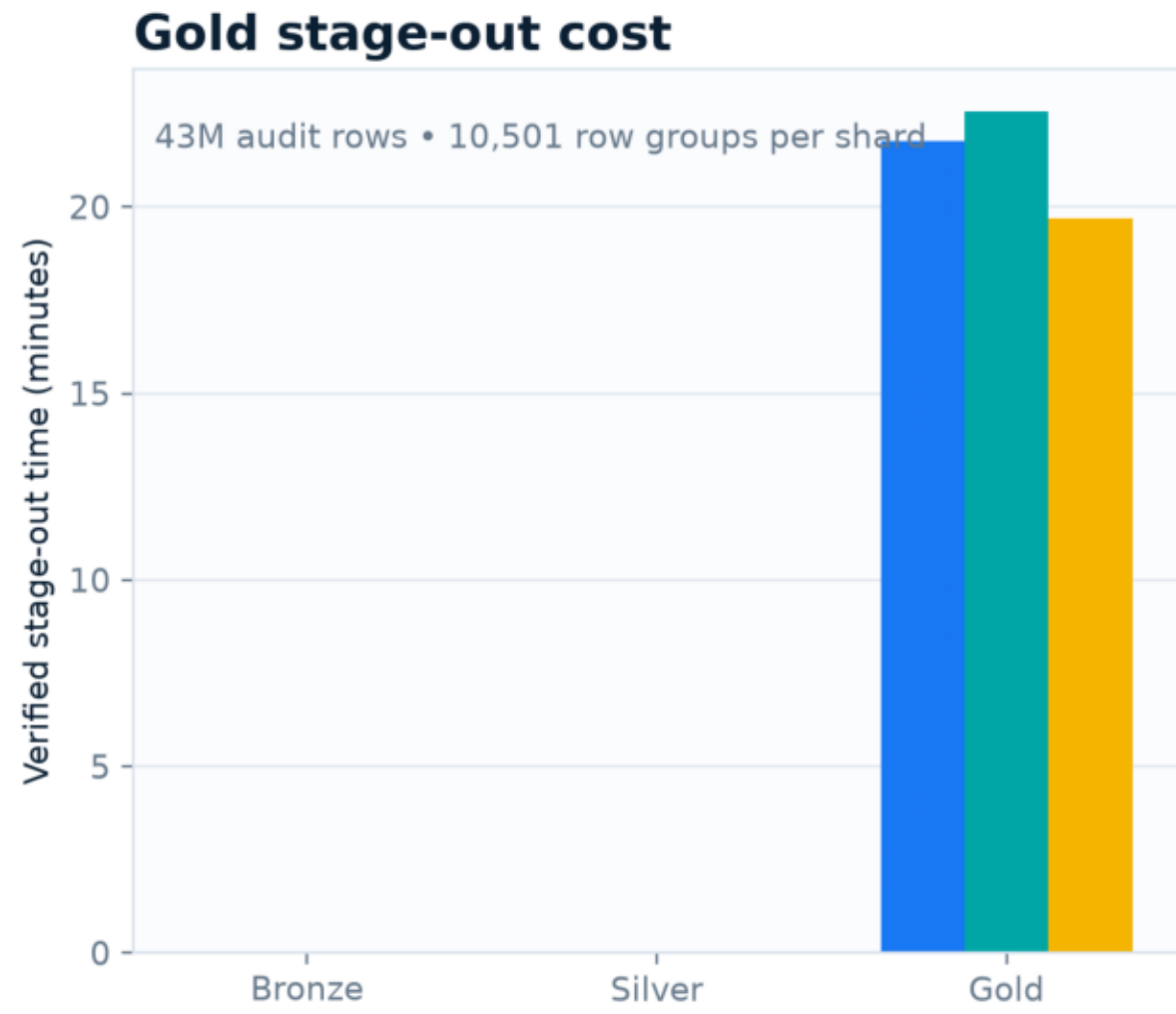
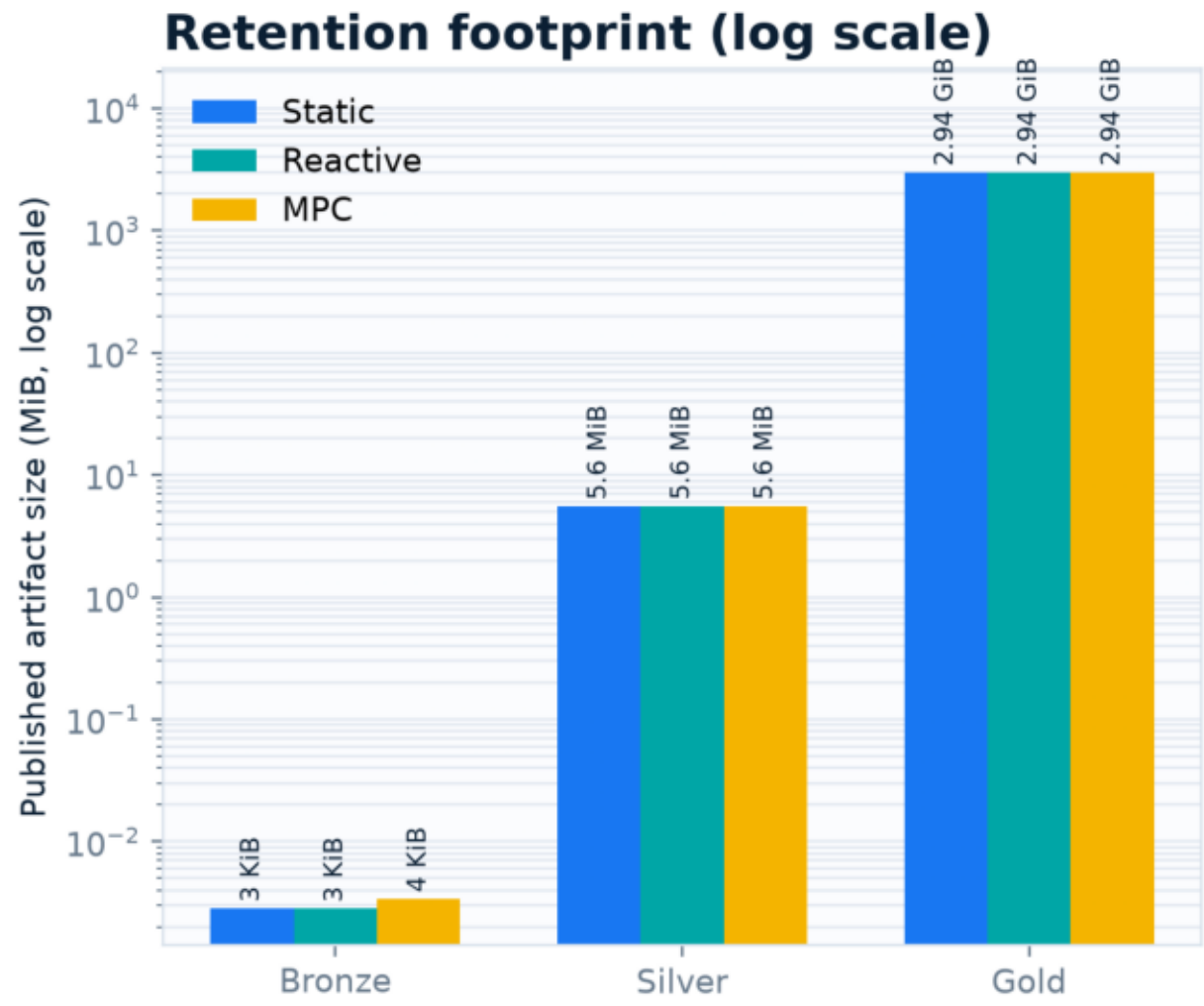


Final production controller outcomes across 128 paired seeds

12-node frozen campaign • bars are medians • whiskers show paired 5th-95th percentiles

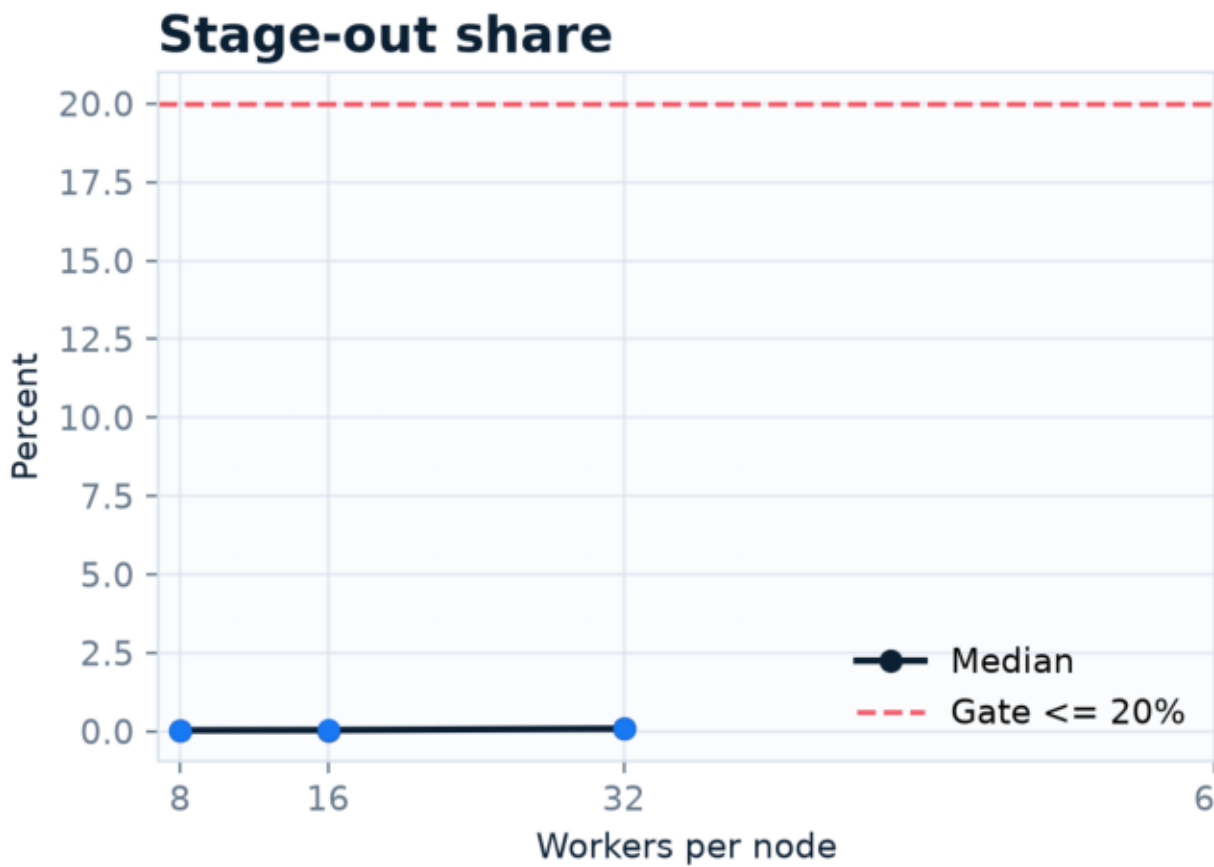
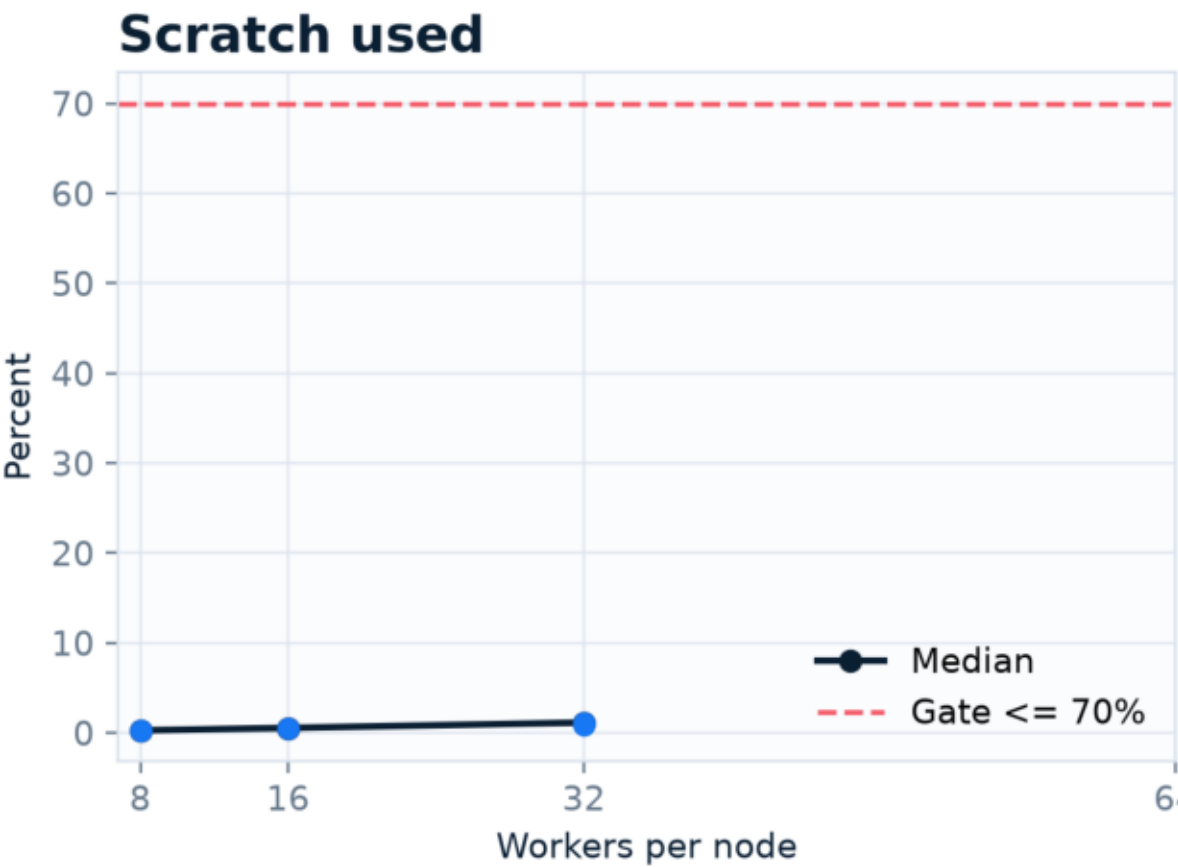
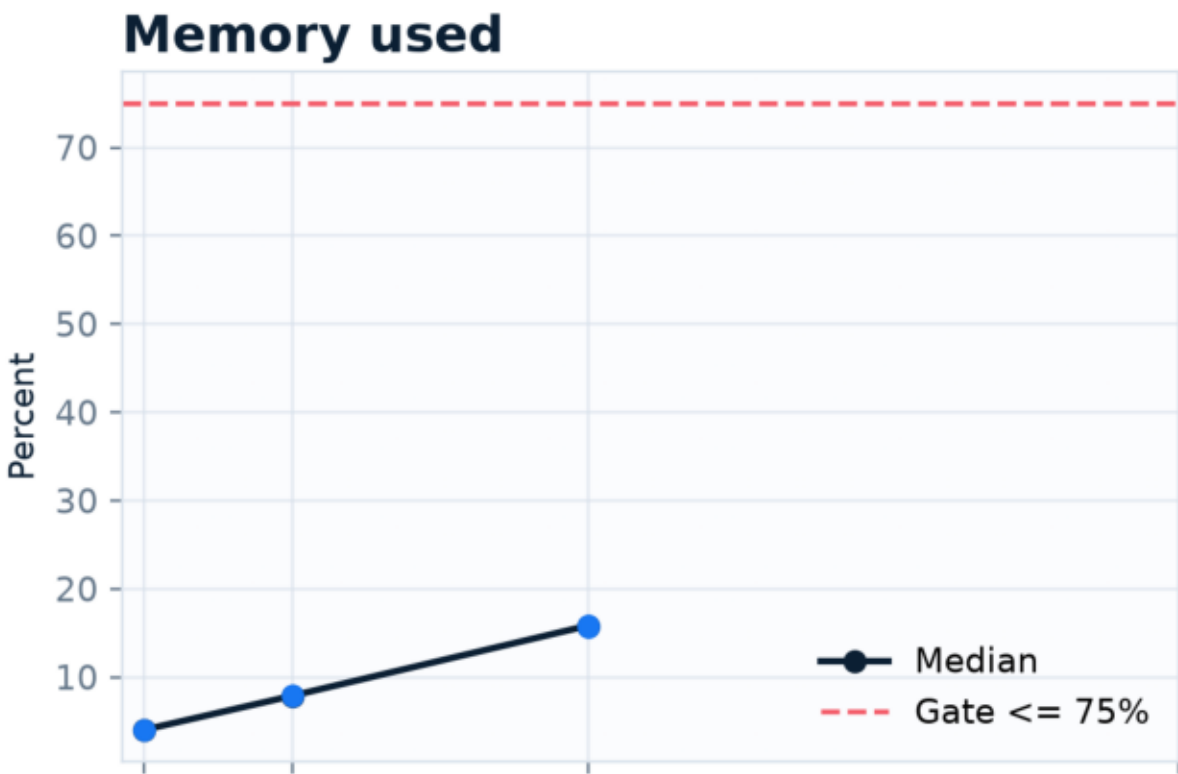
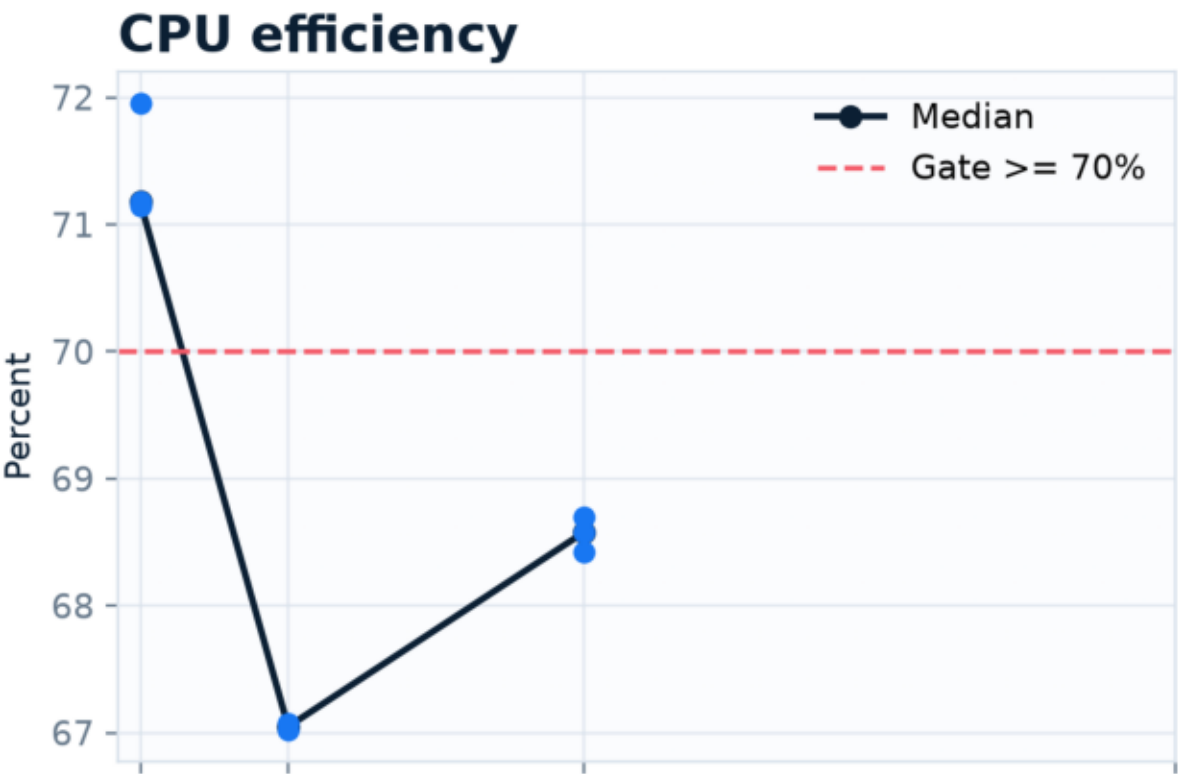


Bronze / Silver / Gold artifact economics



One-node packing ladder — selected rung: 8 workers

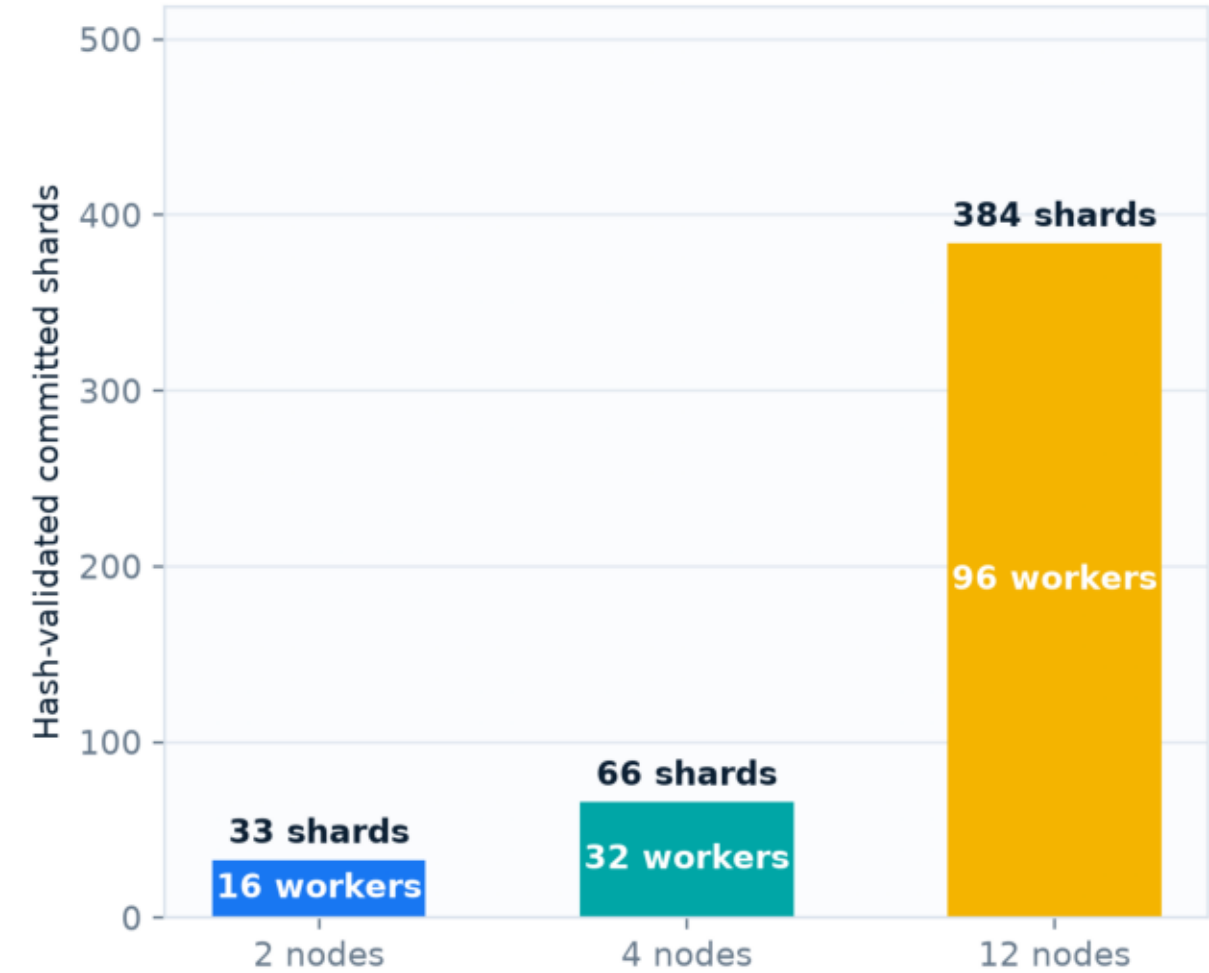
8: 3/3 PASS • 16: 2/3 INCOMPLETE / FAIL • 32: 3/3 FAIL • 64: 0/3 NO FORMAL DATA



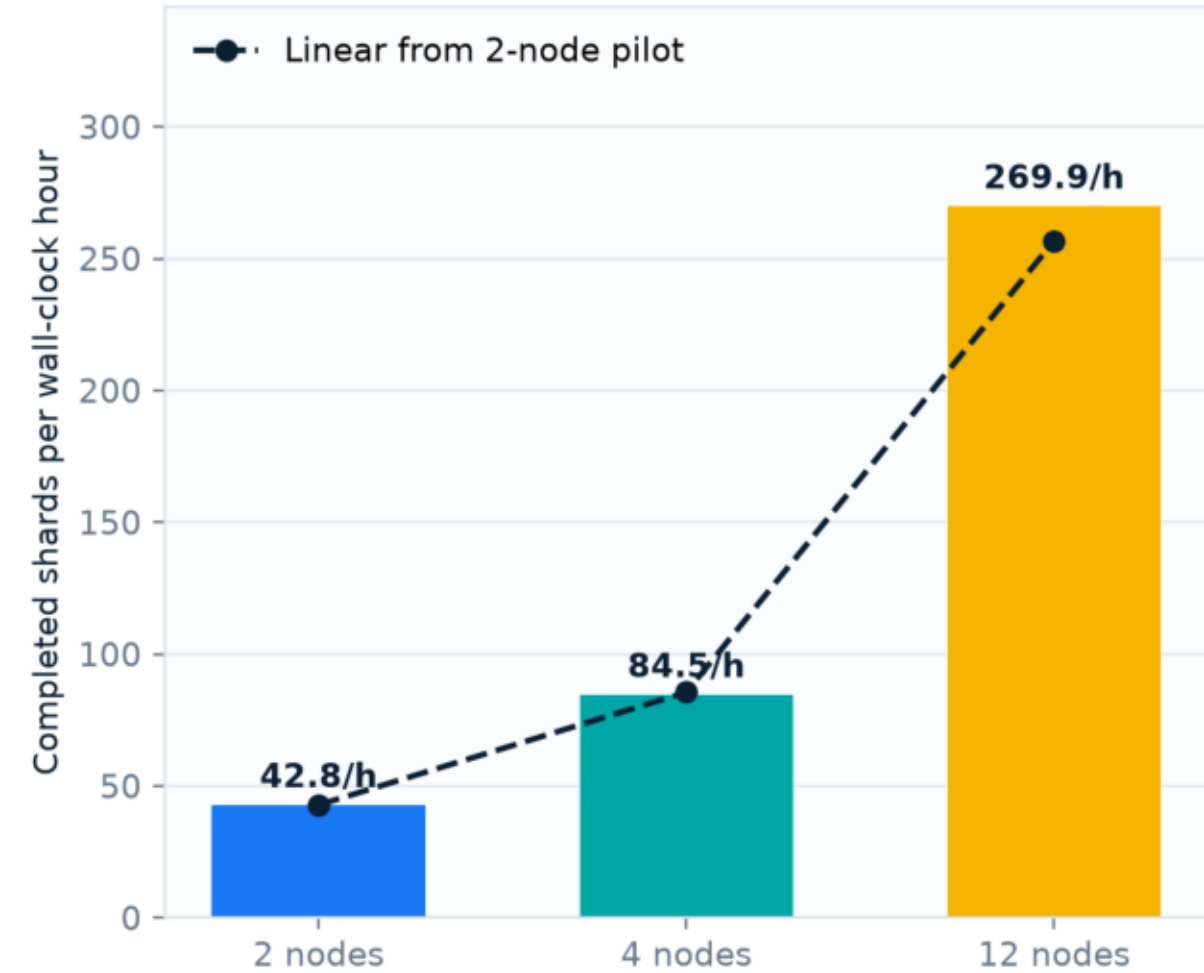
Gated production scales from 2 to 12 nodes

Frozen inputs • balanced mix • zero failures • 12-node run: 4 waves/node; pilots: 2

Committed evidence



Campaign throughput



Production resource gates have substantial headroom

0 failures • 0 swap bytes • maximum stage-out share 0.035%

